

Brutal Series - Paper B58 : Nutrition in Plants, Integers, Motion & Time, The Triangle & its Properties

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A leaf makes 12 mg of food per hour in bright sunlight, but only one-third as much in cloudy light. On a day it gets 5 hours of bright light and 6 hours of cloudy light. How much food does it make in all? [\[Nutrition in Plants\]](#)

- (A) 36 mg
- (B) 84 mg
- (C) 132 mg
- (D) 60 mg
- (E) 24 mg
- (F) 96 mg
- (G) 72 mg

2. Evaluate: $(-7) + 15 - (-8) + (-20)$. [\[Integers\]](#)

- (A) -4
- (B) 4
- (C) -14
- (D) -50
- (E) 20
- (F) -34

3. A car moves steadily at 72 kilometres per hour. Express this speed in metres per second. [\[Motion & Time\]](#)

- (A) 20 m/s
- (B) 26 m/s
- (C) 72 m/s
- (D) 25 m/s
- (E) 200 m/s
- (F) 18 m/s

4. Two angles of a triangle measure 47 degrees and 68 degrees. What is the third angle? [\[The Triangle & its Properties\]](#)

- (A) 75 degrees
- (B) 135 degrees
- (C) 45 degrees
- (D) 25 degrees
- (E) 65 degrees
- (F) 55 degrees
- (G) 115 degrees

5. Pea is a legume. Its root nodules hold bacteria that pull nitrogen from the air and supply 35 kg of nitrogen per hectare free. The pea crop needs 50 kg of nitrogen per hectare in all. How many kg of nitrogen fertiliser must the farmer still add per hectare? [\[Nutrition in Plants\]](#)

- (A) 35 kg
- (B) 85 kg
- (C) 50 kg
- (D) 30 kg
- (E) 15 kg
- (F) 10 kg
- (G) 20 kg

6. Evaluate the product: $(-3) \times (-4) \times (-5)$. [\[Integers\]](#)

- (A) 23
- (B) -12
- (C) -23
- (D) -15
- (E) 60
- (F) -60

7. A scooter covers the first 60 km of a trip at 30 km/h and the next 60 km at 60 km/h. What is its average speed for the whole 120 km trip? [\[Motion & Time\]](#)

- (A) 40 km/h
- (B) 42 km/h
- (C) 50 km/h
- (D) 90 km/h
- (E) 48 km/h
- (F) 45 km/h
- (G) 30 km/h

8. An exterior angle of a triangle is 110 degrees. One of the two interior opposite angles is 45 degrees. What is the OTHER interior opposite angle? [\[The Triangle & its Properties\]](#)

- (A) 55 degrees
- (B) 70 degrees
- (C) 65 degrees
- (D) 110 degrees
- (E) 35 degrees
- (F) 45 degrees
- (G) 155 degrees

9. Through its open stomata a leaf loses 0.5 mL of water per hour by transpiration during the day. At night the stomata are nearly shut and the rate is only one-fifth. Over 8 daytime hours and 10 night-time hours, how much water does the leaf lose? [\[Nutrition in Plants\]](#)

- (A) 1 mL
- (B) 4.5 mL
- (C) 5 mL
- (D) 3 mL
- (E) 5.5 mL
- (F) 4 mL

10. At dawn the temperature is -5 degrees Celsius. It falls 3 degrees each hour for the next 4 hours, then rises 2 degrees each hour for 3 hours. What is the temperature at the end? [\[Integers\]](#)

- (A) -7 degrees Celsius
- (B) -17 degrees Celsius
- (C) -23 degrees Celsius
- (D) 1 degree Celsius
- (E) -11 degrees Celsius
- (F) -5 degrees Celsius
- (G) 11 degrees Celsius

11. A simple pendulum completes 20 full swings (oscillations) in 30 seconds. What is the time period of this pendulum, and how long will 40 swings take? [\[Motion & Time\]](#)

- (A) 3 s per swing; 40 swings take 120 s
- (B) 1.5 s per swing; 40 swings take 45 s
- (C) 0.75 s per swing; 40 swings take 30 s
- (D) 2 s per swing; 40 swings take 80 s
- (E) 1.5 s per swing; 40 swings take 30 s
- (F) 0.67 s per swing; 40 swings take 27 s
- (G) 1.5 s per swing; 40 swings take 60 s

12. The two legs of a right-angled triangle are 9 cm and 12 cm long. Find the length of its hypotenuse. [\[The Triangle & its Properties\]](#)

- (A) 18 cm
- (B) 16 cm
- (C) 21 cm
- (D) 15 cm
- (E) 13 cm
- (F) 11 cm

13. To build one batch of glucose a leaf uses 6 units of carbon dioxide and 6 units of water. A leaf is supplied 30 units of carbon dioxide and only 24 units of water. How many full batches of glucose can it make? [\[Nutrition in Plants\]](#)

- (A) 5 batches
- (B) 3 batches
- (C) 4 batches
- (D) 24 batches
- (E) 6 batches
- (F) 9 batches
- (G) 30 batches

14. Evaluate: $(-48) \div 8 \div (-3)$. [\[Integers\]](#)

- (A) -2
- (B) 18
- (C) 16
- (D) 2
- (E) -6
- (F) 6
- (G) -18

15. A sprinter runs at 25 metres per second for a short burst. Express this speed in kilometres per hour.

[Motion & Time]

- (A) 250 km/h
- (B) 7 km/h
- (C) 45 km/h
- (D) 90 km/h
- (E) 25 km/h
- (F) 9 km/h
- (G) 36 km/h

16. In a right-angled triangle the hypotenuse is 26 cm and one leg is 10 cm. How long is the other leg? [The

Triangle & its Properties]

- (A) 36 cm
- (B) 22 cm
- (C) 20 cm
- (D) 24 cm
- (E) 676 cm
- (F) 18 cm
- (G) 16 cm

17. Bread mould spreads over stale bread, releases juices that break the bread down outside its body, and then soaks up the dissolved food. What is this mode of nutrition called? [Nutrition in Plants]

- (A) Insectivorous nutrition
- (B) Heterotrophic by photosynthesis
- (C) Autotrophic nutrition
- (D) Symbiotic nutrition
- (E) Saprotrophic nutrition
- (F) Holozoic nutrition
- (G) Parasitic nutrition

18. A diver starts at the sea surface (0 m). She goes down 18 m, then up 7 m, then down 11 m, then up 4 m. Taking below-surface as negative, where is she now? [Integers]

- (A) 40 m
- (B) -40 m
- (C) -18 m (18 m below the surface)
- (D) -22 m
- (E) -14 m
- (F) -8 m

19. A cyclist rides at a steady 15 metres per second for 4 minutes. How far does the cyclist travel, in kilometres? [Motion & Time]

- (A) 1.8 km
- (B) 3600 km
- (C) 6 km
- (D) 36 km
- (E) 0.36 km
- (F) 3.6 km
- (G) 60 km

20. An isosceles triangle has a vertex (apex) angle of 40 degrees. What is the size of EACH base angle?

[\[The Triangle & its Properties\]](#)

- (A) 80 degrees
- (B) 20 degrees
- (C) 70 degrees
- (D) 140 degrees
- (E) 50 degrees
- (F) 100 degrees
- (G) 40 degrees

21. Cuscuta (amarbel) is a pale yellow climber that twines around a host plant and pushes tiny suckers into it to draw food. Why is Cuscuta yellow instead of green? [\[Nutrition in Plants\]](#)

- (A) It reflects all green light back
- (B) It makes food only at night so it stays pale
- (C) It has no chlorophyll, so it cannot make its own food and must take it from the host
- (D) It has extra chlorophyll that looks yellow
- (E) Its roots are too deep to reach sunlight
- (F) Sunlight bleaches its green colour each day
- (G) It stores too much starch which turns it yellow

22. Evaluate using order of operations: $-20 + (-6) \times (-3) - 12 \div (-4)$. [\[Integers\]](#)

- (A) 41
- (B) -2
- (C) 11
- (D) -35
- (E) -41
- (F) 1

23. A car's odometer reads 56789 km at the start of a journey and 57104 km at the end. The trip took 5 hours. What was the car's average speed? [\[Motion & Time\]](#)

- (A) 60 km/h
- (B) 62 km/h
- (C) 315 km/h
- (D) 53 km/h
- (E) 63 km/h
- (F) 45 km/h
- (G) 1575 km/h

24. In an isosceles triangle each base angle measures 50 degrees. What is the vertex (apex) angle? [\[The](#)

[Triangle & its Properties\]](#)

- (A) 90 degrees
- (B) 80 degrees
- (C) 65 degrees
- (D) 40 degrees
- (E) 130 degrees
- (F) 100 degrees

- 25.** A pitcher plant grows in soil that is very poor in nitrogen. It traps and digests insects inside its pitcher. Which nutrient is the plant mainly getting from the insects? [\[Nutrition in Plants\]](#)
- (A) Oxygen
 - (B) Chlorophyll
 - (C) Sunlight
 - (D) Water
 - (E) Nitrogen
 - (F) Carbon dioxide
 - (G) Glucose
- 26.** In a quiz, every correct answer gives +5, every wrong answer gives -2, and an unattempted question gives 0. A student answers 12 correctly, 5 wrongly, and leaves 3 unattempted. What is the score? [\[Integers\]](#)
- (A) -10
 - (B) 70
 - (C) 55
 - (D) 60
 - (E) 45
 - (F) 50
- 27.** Three vehicles move at these speeds: a bus at 36 km/h, a bike at 12 m/s, and a tractor at 0.5 km per minute. Which is the FASTEST? [\[Motion & Time\]](#)
- (A) The tractor at 0.5 km per minute
 - (B) Cannot be compared
 - (C) They are all equal
 - (D) The bus and tractor are tied fastest
 - (E) The bus at 36 km/h
 - (F) The bike at 12 m/s
- 28.** In a triangle the three angles have the ratio 2 to 3 to 4. Find the biggest of these angles. [\[The Triangle & its Properties\]](#)
- (A) 60 degrees
 - (B) 120 degrees
 - (C) 100 degrees
 - (D) 90 degrees
 - (E) 72 degrees
 - (F) 80 degrees
 - (G) 40 degrees
- 29.** A lichen is made of an alga and a fungus living together. The alga makes food by photosynthesis while the fungus provides water, minerals and shelter. What is this living-together relationship called? [\[Nutrition in Plants\]](#)
- (A) Autotrophic relationship
 - (B) Predatory relationship
 - (C) Saprotrophic relationship
 - (D) Holozoic relationship
 - (E) Parasitic relationship
 - (F) Symbiotic relationship (mutual benefit)
 - (G) Insectivorous relationship

30. Use the distributive idea to evaluate quickly: $(-25) \times 17 + (-25) \times 3$. [\[Integers\]](#)

- (A) -350
- (B) -20
- (C) 500
- (D) -500
- (E) -525
- (F) 350

31. On a distance-time graph for a moving object, the line is perfectly horizontal (flat) for a stretch of time. What does this flat part tell us about the object? [\[Motion & Time\]](#)

- (A) The object is slowing down
- (B) The object moves backward
- (C) The object moves at constant high speed
- (D) The object moves at its fastest
- (E) The object is at rest during that time
- (F) The object has the greatest acceleration
- (G) The object is speeding up

32. Two sides of a triangle are 5 cm and 9 cm. Which of the following could be the length of the third side?

[\[The Triangle & its Properties\]](#)

- (A) 3 cm
- (B) 27 cm
- (C) 2 cm
- (D) 4 cm
- (E) 14 cm
- (F) 15 cm
- (G) 13 cm

33. A green leaf is partly covered with black paper, then the whole plant is kept in bright sunlight for a few hours. The leaf is later boiled, decolourised and tested with iodine. Which part turns blue-black? [\[Nutrition in](#)

[Plants\]](#)

- (A) Only the covered part's border
- (B) Only the edges of the leaf
- (C) The whole leaf equally
- (D) Only the uncovered part, because only it could make starch
- (E) Only the covered part
- (F) The covered part more strongly

34. Find the product of all the integers from -3 to 3 (that is, -3, -2, -1, 0, 1, 2, 3). [\[Integers\]](#)

- (A) 36
- (B) 720
- (C) -36
- (D) 6
- (E) -720
- (F) 1
- (G) 0

35. Two pendulums hang side by side. Pendulum A has a longer thread than pendulum B. Which pendulum has the LONGER time period (takes more time for one swing)? [\[Motion & Time\]](#)

- (A) The heavier bob decides, not length
- (B) Both have the same period
- (C) Neither swings at all
- (D) Pendulum B, because shorter is slower
- (E) Pendulum A, because a longer thread gives a longer time period
- (F) Pendulum A only if pushed harder

36. Each interior angle of an equilateral triangle is 60 degrees. What is the size of each EXTERIOR angle?

[\[The Triangle & its Properties\]](#)

- (A) 150 degrees
- (B) 180 degrees
- (C) 30 degrees
- (D) 240 degrees
- (E) 90 degrees
- (F) 120 degrees

37. A variegated leaf (green in the middle, white at the edges) is kept in sunlight and then tested with iodine after removing its colour. Which region turns blue-black? [\[Nutrition in Plants\]](#)

- (A) Only the white edges
- (B) Neither region turns blue-black
- (C) Only the green region, because only it has chlorophyll to make starch
- (D) Only the white region
- (E) The green and white regions equally
- (F) The whole leaf turns blue-black
- (G) The boundary line only

38. Evaluate: $(-15) \times 4 + 80 \div (-16)$. [\[Integers\]](#)

- (A) -75
- (B) 65
- (C) -65
- (D) 55
- (E) -55
- (F) -60
- (G) -70

39. A boy walks 1.2 km in 20 minutes at a steady pace. What is his speed in metres per minute, and in kilometres per hour? [\[Motion & Time\]](#)

- (A) 60 m/min and 3.6 km/h
- (B) 600 m/min and 3.6 km/h
- (C) 60 m/min and 1.2 km/h
- (D) 60 m/min and 6 km/h
- (E) 60 m/min and 36 km/h
- (F) 6 m/min and 0.36 km/h
- (G) 1200 m/min and 3.6 km/h

40. The three angles of a triangle are x , $(x + 20)$ and $(x + 40)$ degrees. Find the value of x and the angles.

[The Triangle & its Properties]

- (A) $x = 20$; angles 20, 40, 60 degrees
- (B) $x = 30$; angles 30, 50, 70 degrees
- (C) $x = 50$; angles 50, 70, 90 degrees
- (D) $x = 40$; angles 40, 80, 120 degrees
- (E) $x = 60$; angles 60, 80, 100 degrees
- (F) $x = 40$; angles 40, 60, 80 degrees

41. On a long sunny day a tree's leaves together make 480 g of glucose. The tree uses three-eighths of it for its own respiration and stores the rest. How many grams of glucose are stored? [Nutrition in Plants]

- (A) 160 g
- (B) 300 g
- (C) 240 g
- (D) 320 g
- (E) 120 g
- (F) 180 g

42. A bank account shows a balance of -350 rupees (overdrawn). The owner deposits 500 rupees and then withdraws 200 rupees. What is the new balance? [Integers]

- (A) -650 rupees
- (B) -50 rupees
- (C) 650 rupees
- (D) 50 rupees
- (E) 150 rupees
- (F) -1050 rupees

43. A passenger train moving at a steady 90 km/h must cover a 1.5 km long bridge. How many seconds does the train's front take to cross from one end of the bridge to the other? [Motion & Time]

- (A) 6 s
- (B) 54 s
- (C) 100 s
- (D) 60 s
- (E) 135 s
- (F) 90 s
- (G) 1.5 s

44. A right-angled isosceles triangle has its right angle at one vertex. What are its three angles? [The Triangle & its Properties]

- (A) 90, 60, 30 degrees
- (B) 90, 45, 45 degrees
- (C) 60, 60, 60 degrees
- (D) 90, 50, 40 degrees
- (E) 90, 70, 20 degrees
- (F) 90, 80, 10 degrees
- (G) 90, 45, 55 degrees

45. One leaf absorbs 5 mg of carbon dioxide per hour during photosynthesis. A plant has 240 such leaves and receives 9 hours of good light. How many grams of carbon dioxide does the whole plant absorb that day? [\[Nutrition in Plants\]](#)

- (A) 10.8 g
- (B) 1080 g
- (C) 21.6 g
- (D) 10800 g
- (E) 108 g
- (F) 1.08 g
- (G) 2.16 g

46. Evaluate: $(-6) \times (-6) - (-6)$. [\[Integers\]](#)

- (A) -42
- (B) -30
- (C) 42
- (D) 30
- (E) -48
- (F) 36
- (G) 48

47. A cyclist rides 6 km in 18 minutes, rests for 12 minutes, then rides another 4 km in 10 minutes. What is the average speed for the WHOLE journey, including the rest? [\[Motion & Time\]](#)

- (A) 12 km/h
- (B) 24 km/h
- (C) 20 km/h
- (D) 18 km/h
- (E) 21.4 km/h
- (F) 15 km/h
- (G) 10 km/h

48. In a right-angled triangle the hypotenuse is 17 cm and one leg is 8 cm. Find the other leg. [\[The Triangle & its Properties\]](#)

- (A) 13 cm
- (B) 11 cm
- (C) 12 cm
- (D) 289 cm
- (E) 15 cm
- (F) 25 cm
- (G) 9 cm

49. In photosynthesis the number of oxygen molecules released equals the number of carbon dioxide molecules used, and every 6 carbon dioxide molecules build 1 glucose molecule. If a plant uses 3000 carbon dioxide molecules, how many oxygen and how many glucose molecules form? [\[Nutrition in Plants\]](#)

- (A) 6000 oxygen and 500 glucose
- (B) 500 oxygen and 500 glucose
- (C) 3000 oxygen and 500 glucose
- (D) 3000 oxygen and 600 glucose
- (E) 1500 oxygen and 500 glucose
- (F) 3000 oxygen and 18000 glucose
- (G) 500 oxygen and 3000 glucose

50. The temperature of a freezer is -12 degrees Celsius and the temperature of a warm room is 7 degrees Celsius. By how many degrees is the room warmer than the freezer? [\[Integers\]](#)

- (A) -19 degrees
- (B) 5 degrees
- (C) 19 degrees
- (D) -5 degrees
- (E) 7 degrees
- (F) 12 degrees
- (G) 26 degrees

51. A pendulum makes 5 complete swings every 10 seconds. What is its frequency (swings per second) and its time period? [\[Motion & Time\]](#)

- (A) 5 swings per second; period 2 s
- (B) 0.5 swings per second; period 0.5 s
- (C) 0.2 swings per second; period 5 s
- (D) 1 swing per second; period 1 s
- (E) 2 swings per second; period 0.5 s
- (F) 10 swings per second; period 0.1 s
- (G) 0.5 swings per second; period 2 s

52. Two exterior angles of a triangle (one at each of two vertices) are 120 degrees and 130 degrees. Find the interior angle at the THIRD vertex. [\[The Triangle & its Properties\]](#)

- (A) 60 degrees
- (B) 40 degrees
- (C) 70 degrees
- (D) 80 degrees
- (E) 110 degrees
- (F) 130 degrees
- (G) 50 degrees

53. Plants need nitrogen to build proteins, but they cannot use the nitrogen gas of the air directly. In nature, the soil's nitrogen is mainly restored by which organism living in the roots of legumes? [\[Nutrition in Plants\]](#)
- (A) Rhizobium bacteria
 - (B) Fungi on dead leaves
 - (C) Honey bees
 - (D) Green algae
 - (E) Cuscuta
 - (F) Earthworms
 - (G) Pitcher plants
54. Find the value of the missing integer x in: $x + (-7) = -3$. [\[Integers\]](#)
- (A) 7
 - (B) 10
 - (C) -3
 - (D) 4
 - (E) 3
 - (F) -10
 - (G) -4
55. A bus travels from town P to town Q at 40 km/h and returns along the same 120 km road at 60 km/h. What is its average speed for the whole there-and-back trip? [\[Motion & Time\]](#)
- (A) 24 km/h
 - (B) 100 km/h
 - (C) 45 km/h
 - (D) 48 km/h
 - (E) 55 km/h
 - (F) 50 km/h
56. A line segment drawn from a vertex of a triangle to the MIDPOINT of the opposite side is called what? [\[The Triangle & its Properties\]](#)
- (A) An altitude
 - (B) An exterior angle
 - (C) An angle bisector
 - (D) The hypotenuse
 - (E) A base
 - (F) A median
57. A farmer's legume field of 6 hectares lets Rhizobium fix 35 kg of nitrogen per hectare each season. If that same nitrogen would otherwise cost 18 rupees per kg as fertiliser, how much money does the farmer save in a season? [\[Nutrition in Plants\]](#)
- (A) Rs 6300
 - (B) Rs 1260
 - (C) Rs 3780
 - (D) Rs 210
 - (E) Rs 3500
 - (F) Rs 1890

58. Find the average of these five integers: -8, -3, 5, 10, -4. [\[Integers\]](#)

- (A) 0
- (B) 2
- (C) -2
- (D) -6
- (E) 5
- (F) 6

59. Which instrument in a vehicle measures the SPEED at which it is moving at this instant, and which measures the TOTAL distance it has travelled? [\[Motion & Time\]](#)

- (A) A clock measures speed; odometer measures distance
- (B) Speedometer measures both
- (C) Thermometer measures speed; speedometer measures distance
- (D) Speedometer measures speed; odometer measures total distance
- (E) Both measure distance
- (F) Both measure speed
- (G) Odometer measures speed; speedometer measures distance

60. In a triangle, angle A is twice angle B, and angle C is three times angle B. Find the three angles. [\[The Triangle & its Properties\]](#)

- (A) B = 40, A = 80, C = 120 degrees
- (B) B = 20, A = 40, C = 60 degrees
- (C) B = 30, A = 60, C = 90 degrees
- (D) B = 25, A = 50, C = 75 degrees
- (E) B = 45, A = 90, C = 135 degrees
- (F) B = 36, A = 72, C = 108 degrees
- (G) B = 30, A = 90, C = 60 degrees

61. A potted green plant is sealed inside a jar and left in complete darkness for a full day. Which gas in the jar will INCREASE? [\[Nutrition in Plants\]](#)

- (A) Nitrogen, because the plant releases it
- (B) Carbon dioxide, because only respiration occurs in the dark
- (C) Chlorophyll gas is released
- (D) Water vapour falls and oxygen rises
- (E) Both oxygen and carbon dioxide rise equally
- (F) Oxygen, because photosynthesis continues
- (G) No gas changes at all

62. Evaluate: $(-144) \div (-12) + (-5) \times 3$. [\[Integers\]](#)

- (A) -3
- (B) -23
- (C) -27
- (D) 27
- (E) 3
- (F) -15
- (G) 23

- 63.** An athlete runs a 400 m track lap in 50 seconds. What is her average speed in m/s, and how far would she go in 2 minutes at this speed? [\[Motion & Time\]](#)
- (A) 6 m/s; 720 m in 2 minutes
 - (B) 8 m/s; 960 m in 2 minutes
 - (C) 8 m/s; 800 m in 2 minutes
 - (D) 8 m/s; 16 m in 2 minutes
 - (E) 8 m/s; 480 m in 2 minutes
 - (F) 20 m/s; 2400 m in 2 minutes
- 64.** A ladder leans against a wall. Its foot is 5 m from the wall and its top reaches 12 m up the wall. How long is the ladder? [\[The Triangle & its Properties\]](#)
- (A) 14 m
 - (B) 7 m
 - (C) 15 m
 - (D) 169 m
 - (E) 13 m
 - (F) 11 m
- 65.** Which of the following is NOT a raw material that a green plant takes IN for photosynthesis? [\[Nutrition in Plants\]](#)
- (A) Water
 - (B) Minerals from soil
 - (C) Oxygen
 - (D) Carbon dioxide
 - (E) Light energy
 - (F) Sunlight (energy)
 - (G) Chlorophyll's help
- 66.** An office lift starts at the ground floor (0). It goes up 9 floors, down 13, up 5, and down 2. The basement levels are counted as negative. Where does the lift end up? [\[Integers\]](#)
- (A) Floor -3
 - (B) Floor -29
 - (C) Floor -1 (one level below ground)
 - (D) Floor 0
 - (E) Floor 29
 - (F) Floor 1
- 67.** A toy car covers equal distances of 4 m in each successive second: 4 m, then 4 m, then 4 m. What kind of motion is this, and what is its speed? [\[Motion & Time\]](#)
- (A) Uniform motion at 4 m/s
 - (B) Uniform motion at 1 m/s
 - (C) Non-uniform motion, speeding up
 - (D) Motion with changing speed
 - (E) Uniform motion at 12 m/s
 - (F) At rest the whole time
 - (G) Non-uniform motion at 4 m/s

68. An exterior angle of a triangle equals $5x$ degrees, and its two interior opposite angles are $2x$ degrees and 60 degrees. Find x and the exterior angle. [\[The Triangle & its Properties\]](#)

- (A) $x = 20$; exterior angle = 40 degrees
- (B) $x = 15$; exterior angle = 75 degrees
- (C) $x = 10$; exterior angle = 50 degrees
- (D) $x = 30$; exterior angle = 150 degrees
- (E) $x = 25$; exterior angle = 125 degrees
- (F) $x = 20$; exterior angle = 100 degrees

69. A leaf has 12000 stomata per square centimetre. A particular leaf has an area of 35 square centimetres. About how many stomata does this leaf have in all? [\[Nutrition in Plants\]](#)

- (A) 42000
- (B) 350000
- (C) 12035
- (D) 42000000
- (E) 4200000
- (F) 420000

70. Which expression has the GREATEST value: $A = (-2) \times (-9)$, $B = (-30) / 2$, $C = -4 - 11$, $D = (-3) \times 5$?

[\[Integers\]](#)

- (A) $B = -15$
- (B) $A = 18$
- (C) $C = -15$
- (D) $B = 15$
- (E) $D = -15$
- (F) $A = -18$
- (G) $D = 15$

71. Light from a lightning flash reaches you almost instantly, but the thunder (sound) travels at about 340 m/s. If you hear the thunder 5 seconds after seeing the flash, how far away did the lightning strike? [\[Motion & Time\]](#)

[\[Motion & Time\]](#)

- (A) 1700 km
- (B) 68 m
- (C) 850 m
- (D) 1700 m
- (E) 1360 m
- (F) 3400 m
- (G) 340 m

72. What is the sum of all three EXTERIOR angles of any triangle (one taken at each vertex)? [\[The Triangle & its Properties\]](#)

[\[The Triangle & its Properties\]](#)

- (A) 540 degrees
- (B) 90 degrees
- (C) 120 degrees
- (D) 720 degrees
- (E) 360 degrees
- (F) 180 degrees
- (G) 270 degrees

73. During the day a leaf does BOTH photosynthesis and respiration. It releases 9 units of oxygen by photosynthesis but uses up 2 units of oxygen in respiration in the same time. What is the NET amount of oxygen the leaf gives to the air? [\[Nutrition in Plants\]](#)

- (A) 11 units released
- (B) 7 units used up
- (C) 2 units released
- (D) 5 units released
- (E) 7 units released
- (F) 9 units released
- (G) 18 units released

74. Evaluate: $5 - [3 - \{ 2 - (-4 + 1) \}]$. [\[Integers\]](#)

- (A) 3
- (B) 11
- (C) 1
- (D) 9
- (E) -1
- (F) 7

75. A motorbike starts a journey when the clock shows 9:40 am and finishes at 11:25 am, covering 105 km. What was its average speed? [\[Motion & Time\]](#)

- (A) 105 km/h
- (B) 60 km/h
- (C) 90 km/h
- (D) 70 km/h
- (E) 52.5 km/h
- (F) 63 km/h
- (G) 45 km/h

76. An equilateral triangle has a perimeter of 27 cm. What is the length of each side, and what is each interior angle? [\[The Triangle & its Properties\]](#)

- (A) Each side 3 cm; each angle 60 degrees
- (B) Each side 27 cm; each angle 60 degrees
- (C) Each side 6 cm; each angle 60 degrees
- (D) Each side 9 cm; each angle 60 degrees
- (E) Each side 9 cm; each angle 45 degrees
- (F) Each side 13.5 cm; each angle 60 degrees

77. A leaf makes 50 mg of glucose in a day. One-fifth of this glucose is used up at once in respiration, and the rest is turned into stored starch, but 10 percent of that mass is lost during the conversion. What mass of starch is finally stored? [\[Nutrition in Plants\]](#)

- (A) 40 mg
- (B) 36 mg
- (C) 50 mg
- (D) 4 mg
- (E) 45 mg
- (F) 44 mg

78. Two integers multiply together to give -36. One of the two integers is 4. Find the other integer. [\[Integers\]](#)

- (A) -9
- (B) -12
- (C) -40
- (D) 12
- (E) 9
- (F) -32
- (G) 32

79. On a distance-time graph, line P is steeper than line Q, and both are straight. Which object is moving faster, and why? [\[Motion & Time\]](#)

- (A) Object Q, because it is less steep
- (B) Cannot tell from steepness
- (C) Object P, because a steeper distance-time line means greater speed
- (D) Object P, but only if the line curves
- (E) Object Q, because steeper means slower
- (F) Both move at the same speed
- (G) Neither is moving

80. In a right-angled triangle one of the non-right angles is 37 degrees. What is the third angle? [\[The Triangle & its Properties\]](#)

- (A) 53 degrees
- (B) 63 degrees
- (C) 143 degrees
- (D) 43 degrees
- (E) 47 degrees
- (F) 90 degrees

81. Organisms that make their own food from simple raw materials using sunlight are called autotrophs. Which one of the following is an autotroph? [\[Nutrition in Plants\]](#)

- (A) A mushroom
- (B) A tapeworm
- (C) A pitcher plant's prey
- (D) Cuscuta (amarbel)
- (E) A green mango leaf
- (F) Bread mould
- (G) A cow

82. Starting at 50, you repeatedly subtract 9. After how many subtractions do you FIRST reach a negative number, and what is that number? [\[Integers\]](#)

- (A) After 5 subtractions, reaching 5
- (B) After 6 subtractions, reaching -4
- (C) After 4 subtractions, reaching 14
- (D) After 7 subtractions, reaching -13
- (E) After 5 subtractions, reaching -4
- (F) After 6 subtractions, reaching -9

83. A train 200 m long moves at 20 m/s and passes a stationary pole. How long does the WHOLE train take to pass the pole? [\[Motion & Time\]](#)

- (A) 200 s
- (B) 40 s
- (C) 0.1 s
- (D) 10 s
- (E) 5 s
- (F) 4000 s

84. An isosceles triangle has two equal sides of 8 cm each and a base of 5 cm. What is its perimeter? [\[The Triangle & its Properties\]](#)

- (A) 16 cm
- (B) 24 cm
- (C) 40 cm
- (D) 18 cm
- (E) 26 cm
- (F) 21 cm
- (G) 13 cm

85. Glucose made in the leaf is largely changed into starch for storage. Which simple test would you use in the lab to show that starch is present in a leaf? [\[Nutrition in Plants\]](#)

- (A) Hold a glowing splint to it
- (B) Add limewater and look for a blue colour
- (C) Add iodine solution and look for a blue-black colour
- (D) Weigh the leaf before and after
- (E) Add blue litmus and look for red
- (F) Add iodine and look for a red colour
- (G) Add limewater and look for milkiness

86. Evaluate: $(-2) \times (-2) \times (-2) \times (-2) \times (-2)$. [\[Integers\]](#)

- (A) 32
- (B) -8
- (C) 10
- (D) -32
- (E) 16
- (F) -10

87. A car's speed reading rises from 0 to higher numbers, then stays steady, then falls back toward 0 as it nears a stop. During the steady part, the motion is best described as: [\[Motion & Time\]](#)

- (A) Speeding up then slowing down
- (B) At rest
- (C) Non-uniform motion, slowing down
- (D) Backward motion
- (E) Uniform motion (constant speed)
- (F) Non-uniform motion, speeding up

88. A right-angled triangle has legs of 6 cm and 8 cm. Find both its perimeter and its area. [\[The Triangle & its Properties\]](#)

- (A) Perimeter 48 cm; area 24 square cm
- (B) Perimeter 24 cm; area 24 square cm
- (C) Perimeter 24 cm; area 30 square cm
- (D) Perimeter 28 cm; area 24 square cm
- (E) Perimeter 24 cm; area 14 square cm
- (F) Perimeter 14 cm; area 24 square cm
- (G) Perimeter 24 cm; area 48 square cm

89. In the photosynthesis word equation, carbon dioxide and water combine in the presence of sunlight and chlorophyll to give glucose and one more product. What is that other product released into the air?

[\[Nutrition in Plants\]](#)

- (A) Oxygen
- (B) Hydrogen
- (C) Nitrogen
- (D) Chlorophyll
- (E) Carbon dioxide
- (F) Starch
- (G) Water vapour only

90. On a number line, point A is at -7 and point B is at 5. A point M is exactly halfway between A and B. What integer does M stand on? [\[Integers\]](#)

- (A) -6
- (B) -1
- (C) 0
- (D) -2
- (E) 2
- (F) 1
- (G) 6

91. A car travels at 50 km/h for 2 hours and then at 80 km/h for 1 hour. What is its average speed over the entire 3-hour journey? [\[Motion & Time\]](#)

- (A) 43.3 km/h
- (B) 60 km/h
- (C) 70 km/h
- (D) 55 km/h
- (E) 65 km/h
- (F) 58 km/h
- (G) 130 km/h

92. Can a triangle have two right angles (two angles of 90 degrees each)? [\[The Triangle & its Properties\]](#)

- (A) Yes, if it is isosceles
- (B) No, because that alone uses up all 180 degrees, leaving nothing for the third angle
- (C) Yes, in a right-angled triangle
- (D) Yes, if the sides are equal
- (E) Yes, if it is very large
- (F) Yes, if the third angle is 0 degrees
- (G) No, but it can have two angles of 100 degrees

93. A scientist measures that a water plant gives off 80 oxygen bubbles per minute in strong light. When the light is dimmed to one-quarter of its strength, the rate falls in the same proportion. How many bubbles per minute are released now, and what does this show? [\[Nutrition in Plants\]](#)

- (A) 60 bubbles, showing a small change
- (B) 20 bubbles, showing respiration has stopped
- (C) 20 bubbles, showing photosynthesis slows when light decreases
- (D) 40 bubbles, showing it halves with light
- (E) 80 bubbles, showing light has no effect
- (F) 320 bubbles, showing dim light speeds it up

94. A submarine is at -250 m (below sea level). It dives at 15 m each minute for 4 minutes, then rises at 20 m each minute for 3 minutes. What is its final depth? [\[Integers\]](#)

- (A) -310 m
- (B) -130 m
- (C) -370 m
- (D) 250 m
- (E) -280 m
- (F) -190 m
- (G) -250 m

95. A pendulum's time period is 2 seconds. How many complete swings will it make in 3 minutes? [\[Motion & Time\]](#)

- (A) 360 swings
- (B) 6 swings
- (C) 30 swings
- (D) 90 swings
- (E) 120 swings
- (F) 45 swings
- (G) 180 swings

96. The angles of a triangle are $(2x + 10)$, $(3x - 20)$ and $(x + 40)$ degrees. Find x and hence the largest angle. [\[The Triangle & its Properties\]](#)

- (A) $x = 25$; largest angle is 55 degrees
- (B) $x = 35$; largest angle is 90 degrees
- (C) $x = 25$; largest angle is 65 degrees
- (D) $x = 25$; largest angle is 80 degrees
- (E) $x = 20$; largest angle is 60 degrees
- (F) $x = 30$; largest angle is 70 degrees
- (G) $x = 25$; largest angle is 60 degrees

97. Why must a leaf be boiled first in water and then in alcohol BEFORE the iodine starch test is carried out? [\[Nutrition in Plants\]](#)

- (A) To turn the starch into sugar
- (B) To add chlorophyll to the leaf
- (C) To kill bacteria living on the leaf
- (D) To add fresh starch to the leaf
- (E) To soften the leaf and remove the green chlorophyll so the iodine colour can be seen clearly
- (F) To make the leaf produce more oxygen

98. The sum of all integers from -10 to 10 (including both) is found, then 7 is added to it. What is the final result? [\[Integers\]](#)

- (A) 7
- (B) 14
- (C) 17
- (D) -7
- (E) 110
- (F) -3
- (G) 0

99. A runner covers 100 m in 12.5 seconds. What is the average speed in metres per second, and in kilometres per hour? [\[Motion & Time\]](#)

- (A) 8 m/s and 80 km/h
- (B) 12.5 m/s and 45 km/h
- (C) 8 m/s and 2.88 km/h
- (D) 8 m/s and 14.4 km/h
- (E) 0.8 m/s and 2.88 km/h
- (F) 80 m/s and 288 km/h
- (G) 8 m/s and 28.8 km/h

100. A triangle has one right angle, and its other two angles are in the ratio 2 : 3. Find these two acute angles. [\[The Triangle & its Properties\]](#)

- (A) 30 and 60 degrees
- (B) 45 and 45 degrees
- (C) 36 and 64 degrees
- (D) 30 and 45 degrees
- (E) 40 and 60 degrees
- (F) 36 and 54 degrees